
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) models are now central to scaling large language models, yet their mechanistic differences from dense networks have gone largely unexplained. Prior work established how dense models use *superposition* to represent more features than dimensions, and how superposition is governed by feature sparsity and feature importance. We show that this lens fails for MoEs: neither feature sparsity nor feature importance produces the discontinuous phase changes seen in dense models, and network sparsity (the ratio of active to total experts) is instead the quantity that characterizes MoE representations. We introduce new metrics for measuring superposition across experts and use them to establish that greater network sparsity yields greater *monosemanticity*. Building on this, we redefine expert specialization in terms of monosemantic feature representation rather than load balancing, and demonstrate that experts naturally organize around coherent feature combinations when initialized appropriately. Together these results show that network sparsity makes MoEs more interpretable without sacrificing performance, directly refuting the common assumption that interpretability and capability are fundamentally at odds.

1 INTRODUCTION

Mixture of Experts (MoEs) have become prevalent in state-of-the-art language models, such as Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), primarily for their computational efficiency and performance gains (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work improve routing (e.g., Expert-Choice routing) and training stability/transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical results further establish that learnable routers discover latent cluster structure in data, explaining why experts specialize (Dikkala et al., 2023). Yet despite this widespread adoption, MoEs remain poorly understood from a mechanistic interpretability perspective—a gap this paper closes.

Interpretability-oriented approaches have sought to make expert behavior more transparent. Yang et al. (2025b) proposes MoE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling the number of experts to enable capability editing via expert activation. These works, however, intervene at the level of architecture and leave the core mechanistic questions untouched: how MoEs represent features, how experts shape superposition, and whether experts specialize without added regularization. Mu & Lin (2025) survey MoE research and single out mechanistic interpretability as the key open challenge. We answer these questions directly.

A fundamental obstacle to interpreting neural networks is superposition: models represent more features than they have dimensions. This lets networks pack many sparse features into fewer neurons, but at a steep cost—individual neurons become polysemantic and resist interpretation.

MoE architectures introduce a decisive new axis to this problem: network sparsity. Unlike dense models, which activate all neurons regardless of input, MoEs activate only a fraction of their total parameters (Shazeer et al., 2017). Whereas dense models must pack many sparse

features into shared neurons, MoEs are free to be selective, dedicating entire experts to specific feature combinations. This freedom, we show, is what reshapes their representations.

We investigate whether (1) MoEs exhibit less superposition than their dense counterparts, (2) there is a discrete phase change in the amount of superposition of a particular feature in MoE experts across its relative importance and overall feature sparsity, as seen in dense models, and (3) we can understand expert specialization through the lens of feature representation rather than just load balancing.

We answer these questions using simple models that extend Elhage et al. (2022)’s framework to MoEs. Our contributions are as follows: (1) we establish that, unlike dense models, MoEs do not exhibit sharp phase changes, showing instead continuous transitions as network sparsity increases; (2) we demonstrate that MoEs consistently exhibit greater monosemanticity (less superposition) than dense models with equivalent active and total parameters, with individual experts representing features far more cleanly; and (3) we introduce an interpretability-focused definition of expert specialization grounded in monosemantic feature representation, and show that experts naturally organize around coherent feature combinations rather than arbitrary load balancing.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis holds that networks represent concepts as directions in activation space (Park et al., 2024), yet the number of interpretable features routinely exceeds the available dimensions. Elhage et al. (2022) formalized this phenomenon as superposition, showing that dense models rely on non-orthogonal feature packing to maximize capacity at the cost of polysemanticity. Superposition is efficient (Scherlis et al., 2025), but it forces the use of complex post-hoc disentanglement methods, such as Sparse Autoencoders, to recover monosemantic features (Bricken et al., 2023). Recent work has examined how data correlations shape superposition (Prieto et al., 2025), and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures and establish that network sparsity—not feature sparsity alone—governs representational strategies.

Interpretability of MoEs. Although MoEs are now the standard for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations lags far behind their dense counterparts. Existing analysis confines itself to macroscopic behaviors—routing stability (Zoph et al., 2022), expert choice statistics (Zhou et al., 2022), or latent cluster discovery (Dikkala et al., 2023). More recent interpretability-focused approaches force specialization through architectural constraints, such as predefined concept routing (Yang et al., 2025b) or scaling expert counts to match vocabulary sizes (Park et al., 2025), but these prioritize capability editing and leave the intrinsic feature geometry unexplained. We close this gap: we analyze how experts shape superposition, establish that MoEs exhibit greater monosemanticity than dense models, and put forward a feature-based definition of expert specialization.

3 BACKGROUND

A primary aim of Mechanistic Interpretability is to reverse engineer neural networks; one established method decomposes model representations into a set of human-interpretable concepts termed ‘features’. These features are taken to be linear: any hidden state h can be written as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ is the direction corresponding to feature i , α_i is the activation strength of this feature (roughly, the degree to which feature i is present in the input), and F is the set of all represented features.

The *superposition hypothesis* states that models represent far more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). To hold many more features than dimensions

in a latent space, the feature vectors in F must be packed so that they are not all orthogonal. When two features have interference $\langle \hat{f}_i, \hat{f}_j \rangle \neq 0$, we say they are in *superposition*. Such superposition is tolerable as long as features are sparsely active (a characteristic of most online text data), but it exacts a real cost in interpretability, because α_i then carries spurious activations unrelated to whether feature i is present in the input.

Monosemanticity is a property of individual neurons: a neuron’s activation corresponds cleanly to a single α_i (i.e., features are basis-aligned). When features are orthogonal (not in superposition) but not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. This paper targets the reduction of superposition rather than the enforcement of basis-alignment. For brevity, when we call models, experts, or features “more monosemantic,” we mean they display less superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are often conceptualized as compositions of dense models, with each expert treated as an independent dense network. Whether experts actually represent features the way dense models do has never been settled. We settle it here by directly comparing how MoEs and dense models differ in superposition.

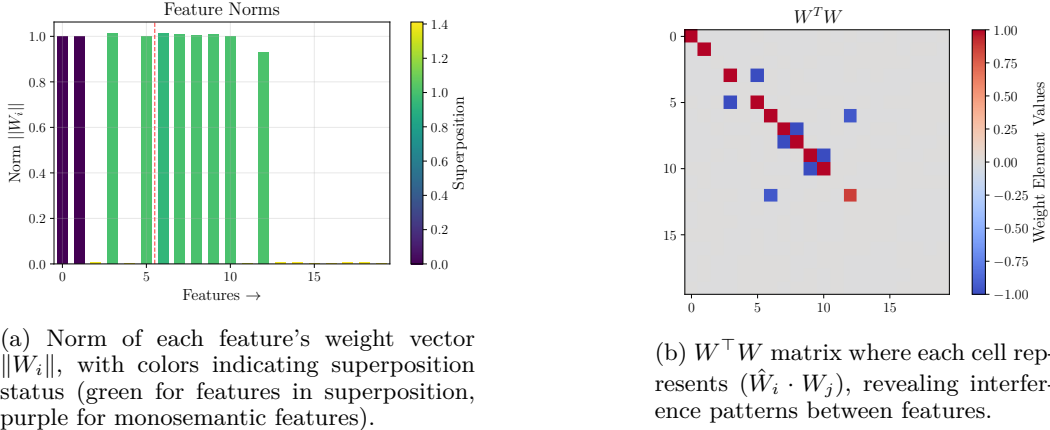


Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, with importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. Superposition (color) is given by $\sum_j (\hat{W}_i \cdot W_j)^2$.

4.1 EXPERIMENTAL SETUP

Our goal is to characterize how a MoE projects a high-dimensional vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and then accurately recovers it. This extends the framework of Elhage et al. (2022) to the MoE setting.

The input distribution. The input vector x represents the activations of an idealized, disentangled model, in which each dimension x_i corresponds to a distinct, independent feature—a perfectly neuron-aligned, monosemantic representation. We take $x_i \sim U(0, 1)$, except that for a given sparsity $S \in [0, 1]$, $P(x_i = 0) = S$. Features do not contribute equally to the loss. To capture this variation in utility, we assign each feature x_i a scalar importance I_i that weights the reconstruction loss. To isolate and study phase transitions for a single feature, we scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, so that $I = (1, 1, \dots, r)$. Together, I_i and r control the signal strength of the features, letting us probe expert sensitivity to feature magnitude directly.

Model Architecture. The MoE consists of E experts, where each expert e is parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. Inputs are assigned to the top- k experts via a learned router $g(x) = \text{softmax}(W^r x)$ where $W^r \in \mathbb{R}^{E \times n}$. Each active

expert projects the input to a lower-dimensional hidden state $h^e = W^e x$ and generates a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^\top h^e + b^e)$ ¹. The final output is the weighted sum of the active experts: $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$ where w_e are the renormalized gating weights.

We train our models with an L2 reconstruction loss weighted by feature importances I_i , given by $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, to prevent expert collapse, we add the standard auxiliary load balancing loss (Fedus et al., 2022), defined as $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the total number of experts, f_e is the fraction of samples in a batch routed to expert e , P_e is the average gating probability assigned to expert e across the batch, and $\alpha = 0.01$ controls the penalty strength. We deliberately omit auxiliary load balancing in Section 4 to isolate the MoE’s intrinsic architectural bias toward or against superposition.

4.2 MEASURING FEATURE CAPACITY

To analyze feature representations across architectures, we compared two fundamental properties of the features: *representation strength* and *interference* with other features. We measured the norm of a feature weight vector in an expert e , $\|W_i^e\|$, which quantifies the extent to which a feature is represented within expert e : $\|W_i^e\| \approx 1$ when feature i is fully represented in expert e and zero when it is not learned. We visualize the interference of a feature i with other features in expert e using the Gram matrix $W^\top W$, whose off-diagonal elements are the pairwise interferences.

As Figures 1a and 2a show, the dense model and the MoE represent a comparable number of features (10 vs 8) with similar norms for equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$). The MoE experts exhibit some local superposition (e.g., Expert 0 in Fig 2b), yet the global interference structure is strictly partitioned. Where the dense model (Figure 1b) allows any feature to interfere with any other, the MoE enforces a block-diagonal structure in which features routed to different experts have exactly zero interference. MoEs therefore allocate representational capacity by partitioning the feature space, sharply reducing the global scope of interference.

Expert Feature Dimensionality. To pin down how MoEs allocate their limited representation capacity differently from the dense model, we measured feature dimensionality—the “fraction of a dimension” that a specific feature receives in a model (Elhage et al., 2022). For a feature i , we define its dimensionality in expert e by

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

D_i^e is bounded between zero (not learned) and one (monosemantic). The total capacity for a MoE can thus be defined as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. When features are “efficiently packed” in a model’s representation space, the dimensionalities of all features sum to the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE, the relation becomes $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint: MoEs pack features exactly as efficiently as dense models for the same total parameters.

Features per Dimension & Network Sparsity. Since both dense and MoE models “efficiently pack” features in their representation space, we compared the number of features per dimension across the models. This lets us measure superposition exactly in both architectures and determine how the number of experts in a MoE affects superposition across feature sparsities. A features-per-dimension value greater than one means the features are

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0, 1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^\top W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

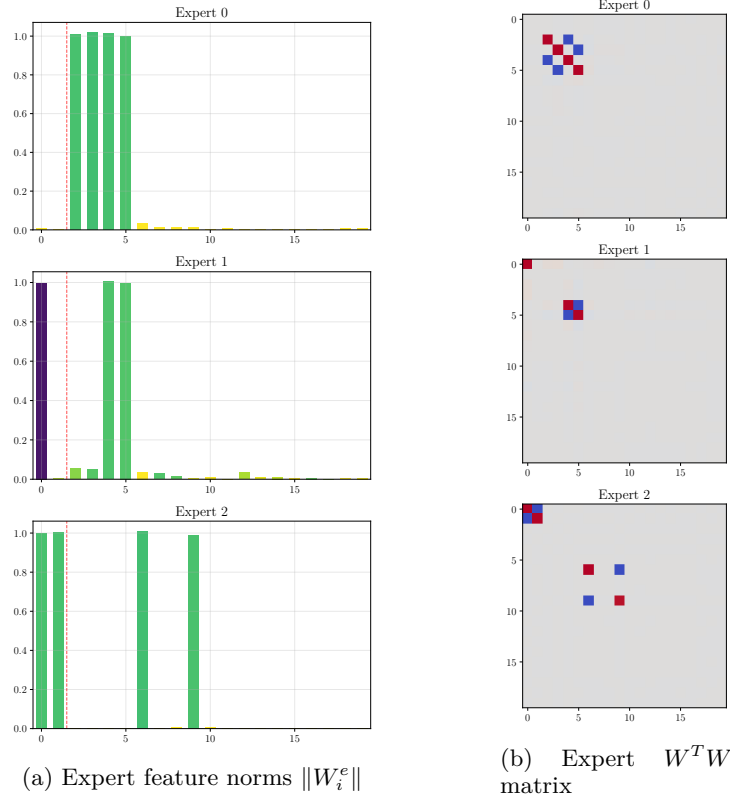


Figure 2: Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert (top- $k = 1$ routing), with importance $I = 0.7^i$ and feature density $1 - S = 0.1$.

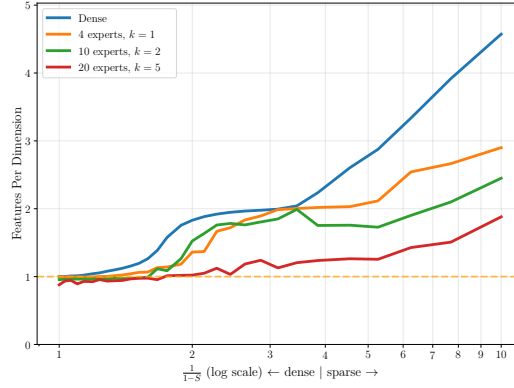


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) has the most superposition, which decreases with increasing expert count: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); 20 experts with $m = 1$, $k = 5$ (red). All models have equal total parameters and similar k/E . The dashed line at 1.0 marks monosemantic representation.

in superposition, since the model is representing more features than it has dimensions. We define features per dimension for a MoE by

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the Frobenius norm and p_e is the expected probability that expert e is used across a batch of input samples, i.e. the average renormalized gating weight after top- k routing.

For MoEs with an *equal* number of total parameters as the dense model, the dense model has a higher number of features per dimension (Figure 3), i.e. more superposition. The dense model exploits superposition to represent a greater total number of features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE caps its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). Moreover, as we increase the total number of experts in the MoE—holding the total parameters and the ratio k/E roughly fixed—the number of features per dimension falls, i.e. superposition drops. *The greater the number of experts, the less superposition in the model.* Features become monotonically more monosemantic as the number of experts grows. This added superposition does buy the dense model consistently lower reconstruction loss than the MoEs (Figure 6, Appendix A.1), with a loss gap at any given sparsity of only ~ 0.03 – 0.08 ; and as the number of experts increases, the MoEs close this gap and match the dense models. See Appendix A.2 for the theoretical intuition.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute them across experts with far less interference. Network sparsity thus reshapes how features are allocated rather than how many are learned. We now examine how properties of the input distribution—feature sparsity and importance—drive this allocation, and whether they induce the physics-inspired *phase changes* seen in dense models.

Models have only a finite set of ways to represent a feature: it may be ignored, superimposed, or monosemantic. A phase change is the appearance of discrete boundaries between these regimes as functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). Varying the sparsity and relative importance of features in the input distribution elicits distinct behaviors; for example, greater feature sparsity drives greater superposition. Analyzing the phase diagram of each expert in a MoE reveals that experts employ fundamentally different representational strategies from dense models.

Setup. We follow the setup of Section 4.1, adding a load balancing loss to prevent specialization collapse—the failure mode in which experts and routers settle into local minima that leave certain experts entirely ignored. We use three model setups, all with one active expert ($k = 1$): (A) $n=2, m=1$; (B) $n=3, m=1$; and (C) $n=3, m=2$. We report the expert-specific phase diagram across feature sparsity and last-feature relative importance for varying network sparsity, increasing the number of experts (E) up to the number of input feature dimensions (n).

In this section we fix active parameters rather than total parameters, so that $m_{\text{dense}} = km_{\text{moe}}$ (ignoring router parameters). This lets us compare within model architectures; otherwise any observed difference could be attributed to architectural changes rather than the number of active parameters. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2. The latter, however, has only one hidden dimension ($m = 1$) to encode the same number of input features ($n = 3$) as the former with two hidden dimensions ($m = 2$), which makes it difficult to use superposition to understand specialization.

Results and Takeaways. In every single-expert (dense) case we observed a clear phase change (Figure 4.X.1/1), confirming Elhage et al. (2022). As soon as we increased the total number of experts, the discrete phase changes vanished. Some experts in MoEs with $E = 2$ echo their respective dense cases (Figure 4.X.2/2) but transition continuously rather than sharply. In each case, one expert became more monosemantic, specializing in the feature

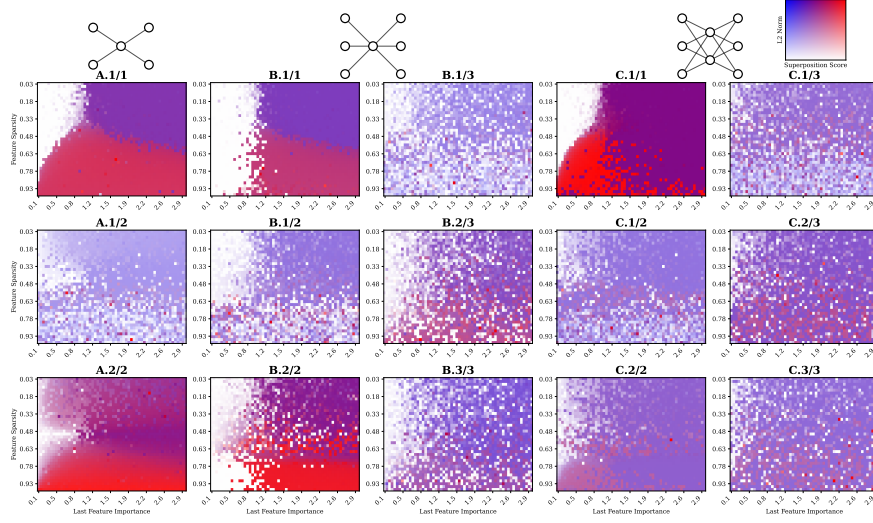


Figure 4: For a particular expert and input dimension (feature), we can decode how it is embedded in the hidden dimension—whether it is ignored (white), monosemantic (blue-purple), or superimposed (red). We plot joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) across varying feature sparsity $S \in [0.1, 1]$ and relative last feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last feature of n total features. For each cell, we train ten models and select the one with the lowest loss. We used load balancing loss in this section. We plot joint feature norm and superposition for the last feature: low L2 norm ($\|W_n\|$) is white, denoting the model is ignoring the last feature; otherwise a low superposition score is blue-purple to indicate monosemantic representation of the last feature. Red indicates the feature is represented in superposition. Cell (i, j) in subfigure X.e/E denotes the expert e of E total experts trained on architecture X for last feature importance $I_n = i$ and sparsity $S = j$; X.1/1 indicates a dense model.

with the highest relative importance. Experts that diverge from the dense cases uniformly show much lower superposition scores (they are bluer), i.e. more monosemantic representations. This is exactly what the previous section predicts: MoEs favor lower superposition than their dense counterparts.

For the $n = 2, m = 1$ setup (Figure 4.A), the dense model fails to represent the last feature when feature sparsity is low, whereas the comparable MoE preserves it far more because it has the capacity to do so. With three input dimensions (Figure 4.B), the MoEs no longer show this behavior: the experts superimpose the other two features, leaving no room for the third feature within a single hidden dimension. Unlike the other two cases, in architecture B the single superimposed hidden dimension is insufficient, in the high-sparsity regime, to represent all features. Even so, no clear phase change appears—except in the 0.5–0.7 feature-sparsity region of 4.B.2/2, which is mostly discrete but mixed. For $m = 2$, the white region of the dense model (Figure 4.C.1/1), in the mid- to low-feature-sparsity domain where the last feature is less important than the others, ignores that feature. As network sparsity increases—across all other panels of Figure 4.C—the models represent the last feature with progressively greater L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). The dimensionality in the low relative-importance region thus rises with network sparsity, exactly as Figure 3 shows.

We observed a window of feature sparsity from roughly 0.48 to 0.7 in Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 with a heavy mix of polysemanticity, monosemanticity, and ignorance. This reveals a middle ground in MoEs where polysemantic and monosemantic representations achieve comparable loss, so the network does not commit cleanly to the strategies seen in the low- and high-sparsity domains. Dense models show no such pattern—direct evidence that MoEs learn genuinely different representational strategies.

Conclusion. These experiments are all top- $k = 1$, so only one expert is active at a time. Even under this most dense-like setting, we see vastly different behavior already in the $E = 2$ case—including when the superimposed hidden dimension has ample capacity to represent all features. We therefore conclude that it is misleading to think of MoEs as aggregations of dense models. The router, by allowing each expert to observe only a subset of the feature domain, fundamentally alters how the experts behave and learn.

6 EXPERT SPECIALIZATION

Having established that MoEs exhibit less superposition, we now examine how these monosemantic features are organized within experts and how that organization relates to specialization.

Expert specialization in MoEs has traditionally been framed around load balancing between experts across all inputs (Chaudhari et al., 2025). This framing misses the natural intuition of specialization: an expert should be used precisely when the concepts it specializes in are present in the input.

We define an expert as specialized if it *occupies* certain feature directions in the input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated, and that together they drive load balancing across experts.

Because we fix $k = 1$, the feature space partitions into convex cone regions (see Appendix A.3), each routed to a particular expert. By definition, $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained within the cone and s is any positive scalar. If a feature vector $\vec{x}$ is routed to an expert, then every $s\vec{x}$ is routed to that same expert. In this case we say feature x is contained within expert e , and that expert e *occupies* x .

We find empirically that small models distributing the input space across more experts achieve lower loss (see Appendix A.4). This raises a sharp question: does allocating the input space to certain experts confer characteristic properties on those experts? Our definition of expert specialization predicts that this allocation implies *monosemanticity*—a correlation we confirm holds for larger toy models (e.g., $m = 10$).

In models with $n > 2$, we test whether initializing experts to occupy features in the input space causes those experts to represent those features more monosemantically. Separately, we test the converse: for the features an expert chooses to represent monosemantically, does the expert occupy those features in the input space?

When the gate matrix is initialized with ones along the main diagonal, each expert monosemantically represents the single feature it initially occupied—and only that feature (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the remaining experts, initialized over less important features, represented neither those features nor the five most important features they were not initialized over. When we break the ordering of feature importance and randomize the features each expert initially occupies, each expert monosemantically represents only the most important feature(s) it was initialized over (Figure 5c).

There is a strong correlation between the features initially routed to an expert and the features that expert represents monosemantically. Experts, moreover, monosemantically represent only important features: this holds whether we initialize each expert with a single important feature explicitly or hand it a set of features, from which it selects the most important feature itself and represents it monosemantically.

Even under uniform feature importance, where experts place all features in superposition with higher interference, the features an expert initially occupies remain *relatively* more monosemantic, averaging superposition scores 1.03 standard deviations below that expert’s own mean.

We then tested for a correlation between an expert representing certain features monosemantically and that same expert occupying those features in the input. We measure usage statistics when those features are *one of many* active features and when they are the *only*

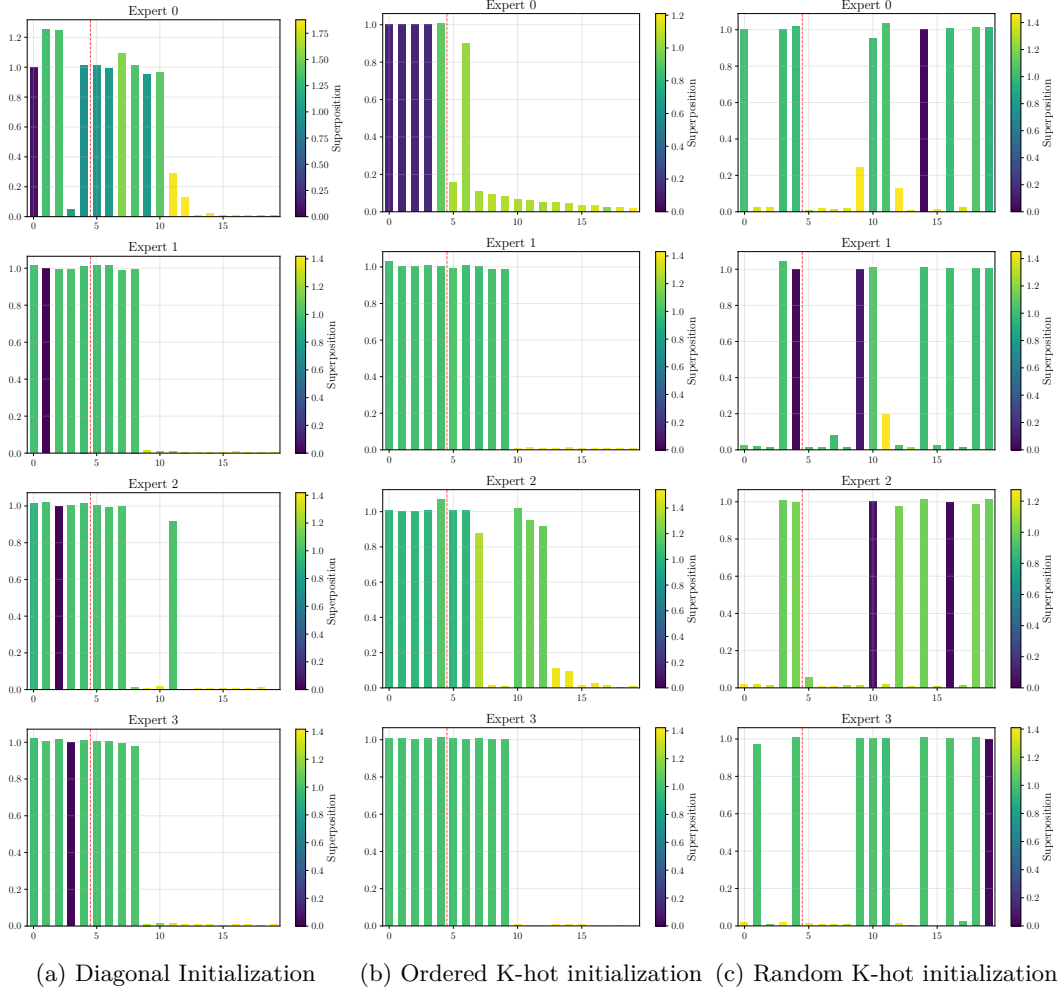


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) results for three different initialization schemes, with $n = 20, m = 5, E = 4, S = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for that dimension), and relative feature importance decreases exponentially in order from feature one to 20. In (b), the gate matrix is initialized to an “ordered k-hot”, such that the first expert aligns with the first five features, and each subsequent expert aligns with the next five features. Relative feature importance is the same as (a). In (c), the gate matrix is initialized to a “random k-hot”, where each expert is assigned five random features such that experts share no common feature but cover all 20 features collectively. Relative feature importance decreases exponentially but is randomly distributed across features.

active features—the latter directly measuring the probability that the expert occupies the features. The correlation holds under both xavier and k-hot initialization (Table 1). With $E = 10$, a mean expert usage of $\sim 10\%$ indicates even load balancing across experts. In every case, when an expert’s monosemantic feature(s) are active, its usage rises sharply; when those features are the only active feature, the expert dominates usage outright. Under k-hot initialization, 100% of all features monosemantically represented by an expert are occupied by that same expert.

The more features an expert represents monosemantically, the more specialized it is. Its usage on arbitrary input falls, but conditional on its specialized features being active, its usage rises far above that of other experts. This holds in every case except the single xavier-initialized expert with four monosemantic features, where utilization drops sharply.

Table 1: Monosemantic feature and usage statistics per expert for $n = 100, m = 10, E = 10$. We train one hundred models per initialization scheme (xavier and k-hot), yielding 1000 experts each. Every statistic is aggregated across models, with experts classified by the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we report expert usage when those feature(s) are one of several active features in the input, and expert usage when those feature(s) are the *only* active feature in the input.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

7 CONCLUSION

We established how experts affect superposition in MoEs, showing that MoEs consistently exhibit greater monosemanticity than dense networks and do not undergo a phase change. We introduced a feature-based definition of expert specialization and demonstrated that experts naturally organize around coherent features when initialization encourages specialization. These findings come from simple autoencoder toy models with synthetic data, so their extension to large-scale transformers with unknown feature distributions remains open (see Appendix A.6). Within these bounds, our results are clear: toy MoEs achieve comparable loss while maintaining markedly more interpretable representations, refuting the prevailing view that mechanistic interpretability and model capability are fundamentally in tension. Future work should identify what favors monosemanticity in MoEs, how MoE training dynamics diverge from dense models, and when specialization emerges. Answering these questions will directly inform the design of more interpretable, high-performing language models.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

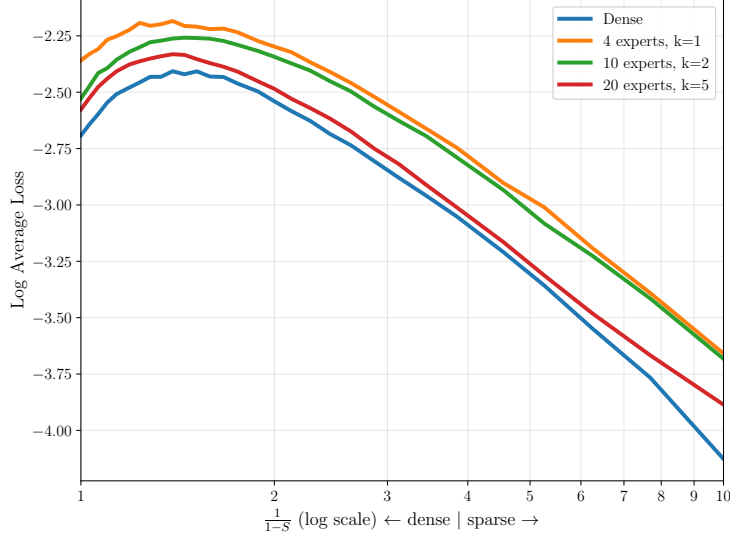


Figure 6: Log average loss versus feature density ($\frac{1}{1-S}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. The dense model outperforms every MoE at every sparsity level, yet the MoE loss converges toward the dense model as the number of experts increases.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs close this same gap structurally, through conditional computation. By aligning activation sparsity with feature sparsity, MoEs eliminate the computational penalty for dedicated, rarely-activating neurons. As Elhage et al. (2022) note, once a model expends computation only on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy—trading the *compression* of superposition for the *selection* of routing.

Geometrically, this structural change manifests as a reduction in *interference*. The global ratio of features represented to parameters stays constant, but the router partitions the feature space so that a feature routed to expert e competes for capacity only with the subset of features also assigned to that expert. Interference is therefore governed by the local expert matrix $(W^e)^\top W^e$ rather than the global $W^\top W$ of a dense model. This partitioning sharply reduces the number of interfering features for any given feature vector, removing the optimization pressure to pack features in superposition and allowing experts to learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the $k = 1$ regime, the router function $g(x) = \text{softmax}(W^r x)$ reduces to $\text{argmax}(W^r x)$. The region routed to expert i is $\forall j \neq i, (w_i - w_j)^\top x > 0$, where w_i and w_j are row vectors of W^r . This is a homogeneous linear inequality, and regions bounded by such inequalities are, by definition, convex cones. If a given x satisfies the inequality, so does sx for any positive scalar s ; and if x_1 and x_2 satisfy it, so does any $x = x_1\lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$.

For $k > 1$, the region of inputs sent to a particular expert e becomes a union of convex cones. A union of convex cones is not, in general, itself a convex cone. The notion of experts occupying feature directions therefore need not hold beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

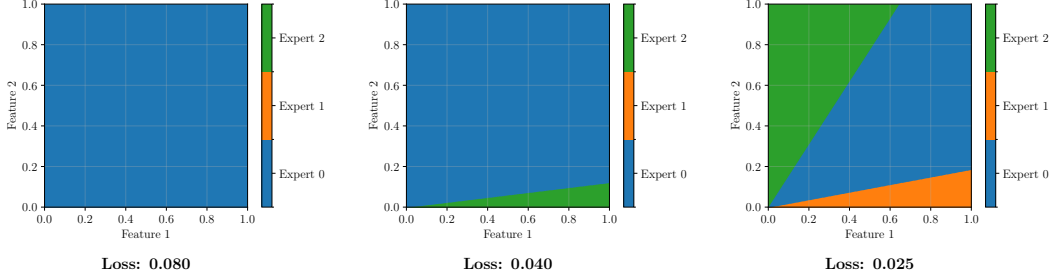


Figure 7: Expert routing of three identical models with differing initialization schemes. We use $n = 2$, $E = 3$, $m = 1$. The first model (left) has the worst performance (loss: 0.08) and routes all inputs to one expert. The second model (middle) has better performance (loss: 0.04) and routes a small portion of inputs, specifically those when feature 1 is active, to a second expert. The third model (right) has the lowest loss (loss: 0.025), and distributes the input space among all experts. One expert is chosen when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2, m = 1, E > 1$), we find empirically that models distributing the input space across more experts achieve lower loss, across $E \in [2, 7]$. Holding $n = 2$ lets us visualize exactly which portions of the input space route to which experts (Figure 7).

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map from any dense model to a monosemantic MoE with an equivalent number of active features under a sparsity constraint.

Assume there exists an upper bound for the number of active features a for any input such that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Furthermore, assume that a is no greater than the hidden dimensionality, m , of an expert, providing an upper bound on the number of features a model has to represent. Assume also that the hidden dimensions is smaller than the total number of input features n ($a \leq m \leq n$). To construct the monosemantic MoE, for each possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —meaning the size of the subset of active features is smaller than or equal to a —create an expert which monosemantically preserves those features. (In fact, you can take only the subsets such that $|S| = a$.) The router then selects the expert which corresponds to those active features (of which there will never be more than a , by assumption):

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has sufficient capacity to represent its assigned features without superposition. To reiterate, only a features are active and every unique combination of active features receives its own dedicated expert with sufficient capacity to represent those features monosemantically. So, the number of possible experts needed is $\binom{n}{m}$.

The reconstruction for this theoretical MoE attains zero loss only while the sparsity constraint holds (or as sparsity goes to one in these toy models), because more than m features may occasionally be active at once ($a \not\leq m$), exceeding the network’s monosemantic representational capacity—though the dense polysemantic model could do no better unless

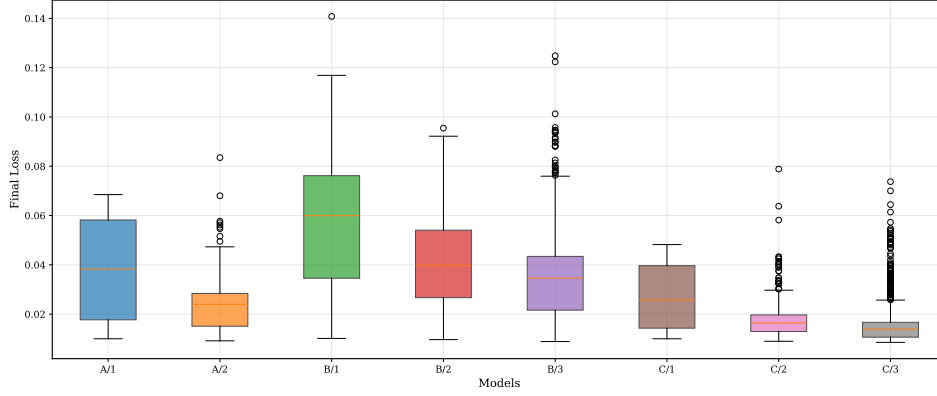


Figure 8: Model X/E uses X to denote the same model architectures and models used in Figure 4 and E denotes the total number of experts (i.e. network sparsity). Increasing network sparsity decreases mean loss while increasing localized variance, most strongly as the number of experts approaches the input feature dimensions. We attribute this variance to the relatively unstable training of MoEs compared to dense models (even though we train ten models per cell and select the lowest loss).

features are correlated in the distribution. Hence, even when $a \not\leq m$, the polysemantic model faces the same obstacle, and the monosemantic MoE under this construction can still outperform it under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ under the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ with $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings of this paper, the sparsity constraint holds in the limit where sparsity goes to one. In practice, moreover, there is likely an upper bound on how many features a given amount of information can semantically encode, set by the size of meaningful embeddings of that data. A MoE with sufficient experts and a tractable, interpretable amount of superposition should therefore suffice to encode all present features.

A.6 LIMITATIONS

Our results should be read in light of several deliberate scope choices. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). This design enables precise measurement of interference and monosemanticity, but it abstracts away complexities of large-scale transformers—multiple computation, attention, heterogeneous feature distributions, and realistic routing dynamics—so transfer to real-world MoE architectures remains to be established.

Second, our architectural and routing choices are intentionally restricted: simple experts, fixed top- k routing (often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs use richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or move beyond reconstruction to next-token prediction, any of which may yield qualitatively different representational behavior.

Third, the feature distributions in our synthetic tasks—sparsity patterns, feature importance, and independence assumptions—are substantially simpler than those in natural data.

Finally, while we establish increased monosemanticity and reduced superposition in MoEs under fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale; prior work shows such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

In sum, this study delivers clear mechanistic insights under clean experimental conditions, and validating these patterns in large, realistic MoE systems is the natural next step. Scaling our methods to larger models is nontrivial and is the clearest direction for future work.